Lab #14B: Visual Basic Programming: Object Detection Part 2

Objectives: Use Visual Basic and TensorFlow to develop programs to train, detect and classify images.

Task #0: Copy the "assets/images" folder and the "images" folder in "Part1", paste them into "Part2/assets" folder, and "Part2" respectively.

Task #1: Open file "Part2.sln" in Part2 folder with Visual Studio 2019. Review the code template and Test webcam. You need to plug in a webcam to test webcam capturing function.

Task #2: Use the excel file "PLC_Link.xlsm" in the USB-drive, copy the path to the file, paste in VB.NET code - "Part2.sln" - Form1.vb, the following line:
Private Const ExcelWorkbookPath As String = "FILE PATH"

Task #3: Review the code template, train the images, and run object detection using webcam.

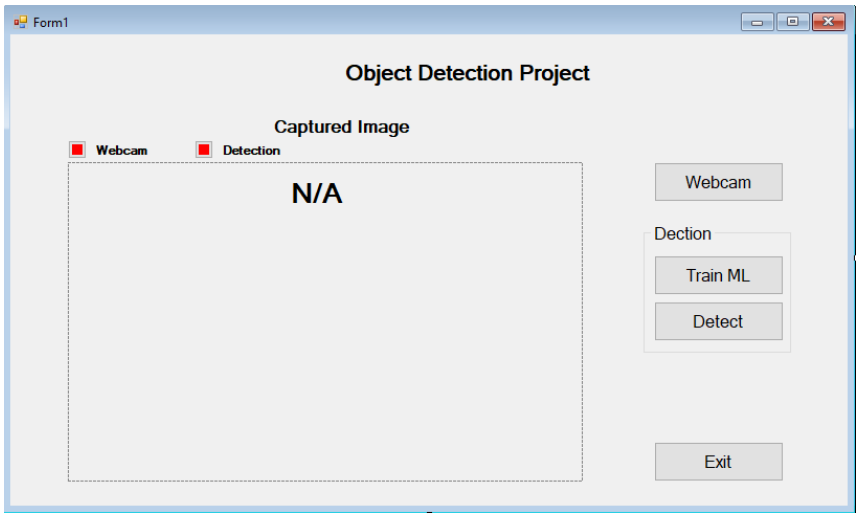

Figure 1. Object detection UI

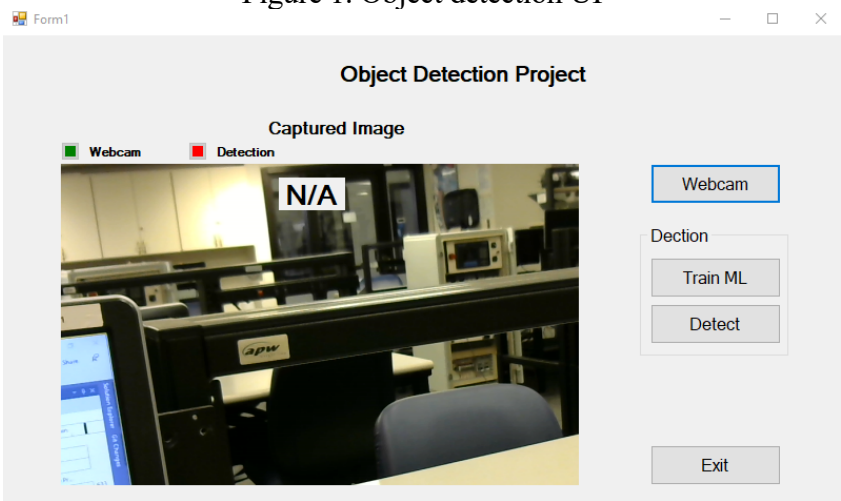

Figure 2. Example of webcam capturing process

Part A: Now we have all integrated image files as the training sample. Run Button “Train ML” to use TensorFlow to train all images based on the giving categories. Check the Visual Studio console or textbox in the design to monitor the training status.

Part B: Run Button “Detect” and hold your webcam to detect object or image. The label on the picturebox will display the detected result using TensorFlow model.

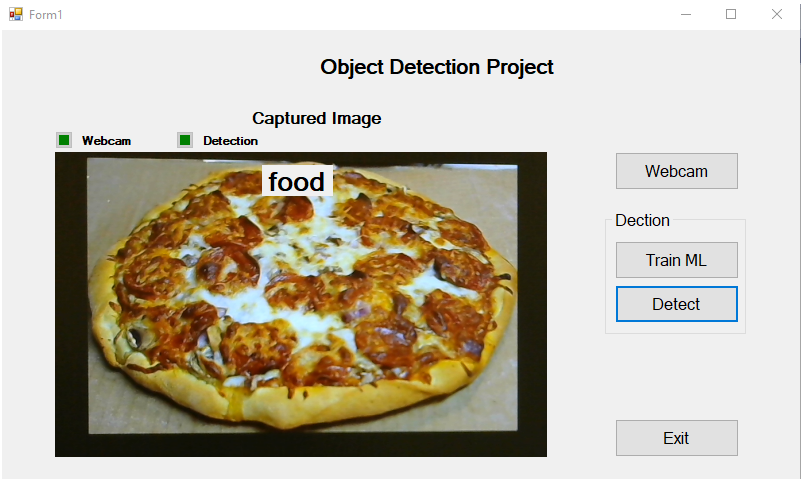

Figure 3. Webcam capturing and object detecting process

Task #4: Re-try Task #2 using different images and categories to test the performance of object detection.

Show your work to your instructor after you are done.